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Advanced Composite Materials 5
Lab Report

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Notation

b – Specimen width
d – Specimen thickness
D – Mid-span deflection
E_1 – Longitudinal elastic modulus
E_2 – Transverse elastic modulus
E_b^x – Flexural modulus
E_f – Fibre modulus
E_m – Matrix modulus
E_x – Off-axis Young's modulus at angle θ
F – Applied load
G_{12} – In-plane shear modulus
G_f – Fibre shear modulus
G_m – Matrix shear modulus
L – Flexural span length
S_6 – In-plane shear strength
X_1^C – Longitudinal compressive strength
X_1^T – Longitudinal tensile strength
X_2^C – Transverse compressive strength
X_2^T – Transverse tensile strength
ε – Normal strain
ε_x – Strains in the longitudinal direction
ε_y – Strain in the transverse direction
ϕ_f – Fibre volume fraction
ν_{12} – Major Poisson's ratio (strain in 2 due to stress in 1)
ν_f – Fibre Poisson's ratio
ν_m – Matrix Poisson's ratio
σ – Normal stress
σ_b^x – Flexural (bending) strength
τ_6 – In-plane shear stress
θ – Fibre orientation angle relative to loading direction

1 Summary

The objective of the testing procedures and results described in this laboratory report was to characterise the tensile, compressive, and flexural response of unidirectional and multidirectional carbon fibre-epoxy composite laminae made from VTC401 epoxy prepreg with T700G fibres. Tests were conducted in accordance to test standards with results compared to micromechanics predictions and data sheet values to assess material behaviour and accuracy to manufacturer specifications.

All laminates exhibited anisotropic behaviour, with mechanical properties strongly dependent on fibre orientation. The longitudinal tensile and compressive properties were fibre-dominated, while the transverse and off-axis responses were matrix and shear-dominated.

Table 1: Summary of measured and theoretical/datasheet values with percentage differences

Property	Symbol	Theoretical / Datasheet	Experimental	Difference [%]	Comment / Comparison
Longitudinal modulus	E_1	133.1 GPa	117.9 GPa	11.4	Slightly lower due to possible lower fibre volume fraction, fibre misalignment and voids.
Transverse modulus	E_2	5.27 GPa	7.3 GPa	38.5	Micromechanics underestimates due to matrix inhomogeneity.
In-plane shear modulus	G_{12}	2.49–4.08 GPa	3.07 GPa	–	Within expected range, consistent with lower bound and lower fibre fraction.
Longitudinal tensile strength	X_1^T	2040 MPa	2366 MPa	16.0	Higher strength; datasheet values may be conservative.
Transverse tensile strength	X_2^T	75 MPa	50.3 MPa	32.9	Lower value due to lower fibre volume fraction and matrix-dominated failure.
In-plane shear strength	S_6	128 MPa	120 MPa	6.3	Slightly lower due to lower fibre volume fraction.
Off-axis modulus (15°)	$E_{x,\theta=15}$	35.7 GPa	35.3 GPa	1.1	Excellent agreement.
Off-axis modulus (30°)	$E_{x,\theta=30}$	13.6 GPa	14.2 GPa	4.4	Minor deviation.
Off-axis modulus (45°)	$E_{x,\theta=45}$	8.58 GPa	7.54 GPa	12.1	Deviation increases due to matrix shear effects.
Longitudinal compressive strength	X_1^C	–	1040 MPa	–	44% of X_1^T due to fibre microbuckling.
Transverse compressive strength	X_2^C	–	127 MPa	–	252% of X_2^T; epoxy stronger in compression than tension.
Flexural modulus (QI laminate)	E_x^b	–	11.5 GPa	+50.7 vs. XP	QI laminate stiffer due to $\pm 45^\circ$ plies, but fails at a lower strain
Flexural strength (QI laminate)	σ_x^b	–	126 MPa	+24.8 vs. XP	Improved load distribution and shear resistance.

Key Findings:

- Experimental elastic and strength properties generally matched predictions within 10–20%, with notable exceptions being the transverse modulus and tensile strength.
- Deviations mainly arise from matrix property assumptions affecting transverse loading results, limitations of the theoretical equations used, and a lower fibre volume fraction than assumed.
- E_1 and G_{12} values were close to micromechanics predictions, validating fibre–matrix interaction assumptions. E_2 was 38.5% higher than the theoretical prediction.
- Off-axis modulus E_x predictions matched experiments up to 30°, diverging at 45° due to matrix shear effects. Tensile strength decreases exponentially off-axis from the longitudinal direction.
- Compressive strength was 44% lower than tensile strength longitudinally, which is consistent with fibre microbuckling theory, that fibres are stronger in tension. The transverse compressive strength was 252% higher than the tensile strength, as the matrix is stronger in compression than in tension.
- Flexural tests showed that quasi-isotropic laminates have greater stiffness (+50.7%) and strength (+24.8%) than cross-ply laminates due to balanced fibre orientations, but experienced complete failure at lower strain, while the cross-ply laminate failed in steps as each lamina layer fractured.
- Failure modes varied with orientation: brittle fibre fracture at 0°, matrix cracking and delamination at higher angles in tension. Fibre micro-buckling occurred under compression.

2 Experimental Procedure

2.1 Laminate Manufacture

Each VTC401 Expoy Component Prepreg ply was cut into 300 × 300 mm squares using a craft knife and manually laid up according to the fibre orientation listed in Table 2. The sheets were joined by applying pressure along the fibre direction using a plastic spreader to ensure good adhesion and remove trapped air, to consolidate the layers and minimise void formation before curing.

After layup, the laminea was covered by a breather cloth, vacuum bagged and debulked to further consolidate the plies and remove residual air pockets, before being placed into an oven for curing. After curing, the specimens were extracted as per the specimen extraction drawings (Appendix A). Target dimensions are listed in Table 2. End tabs were applied to tensile and compression specimens.

Table 2: Layups and specimen information from extraction drawings

Layup	Layers	Test	Dimensions [mm]	Specimens
$[0]_4$	4	0° tension	250 × 15 × 1	10
$[90]_8$	8	90° tension	250 × 5 × 2	10
$[0]_8$	8	0° compression	140 × 13 × 2	10
$[90]_8$	8	90° compression	140 × 13 × 2	10
$[\pm 45]_{2s}$	4	± 45 tension (shear)	250 × 25 × 2	5
$[15]_8$	8	15° tension	250 × 25 × 2	10
$[30]_8$	8	30° tension	250 × 25 × 2	10
$[45]_8$	8	45° tension	250 × 25 × 2	10
$[0/90]_{2s}$	8	Cross-ply flexure	100 × 15 × 2	5
$[0/90/\pm 45]_s$	8	Quasi-isotropic flexure	100 × 15 × 2	5

Dimension measurements were taken of all specimens during testing using a digital vernier caliper with an accuracy of $\pm$0.01 mm. Width and thickness were measured at the top, middle, and bottom of each specimen. These values were averaged to determine the cross-sectional area used for stress calculations. Measured dimensions for each specimen are presented in Table 3.

Table 3: Measured widths and thicknesses of composite specimens. Length dimensions in mm, area is mm^2

Specimen	Width 1	Width 2	Width 3	Avg. Width	Thick. 1	Thick. 2	Thick. 3	Avg. Thick.	Area
UD 0°	14.75	14.98	14.94	14.89	0.92	0.92	0.96	0.93	13.85
UD 90°	25.23	25.20	25.16	25.20	1.72	1.70	1.68	1.70	42.84
PM 45°	24.99	24.98	24.99	24.99	1.81	1.78	1.82	1.80	44.98
15°	24.82	24.77	24.75	24.78	1.91	1.89	1.88	1.89	46.83
30°	24.94	25.12	25.00	25.02	1.71	1.75	1.79	1.75	43.79
45°	25.08	25.07	25.01	25.05	1.95	1.93	1.94	1.94	48.60
Compression 0°	13.16	13.10	13.09	13.12	1.70	1.73	1.71	1.71	22.57
Compression 90°	13.37	13.23	13.44	13.35	1.77	1.81	1.78	1.79	23.90
Flexure XP	15.09	15.16	15.10	15.12	1.80	1.77	1.76	1.78	26.91
Flexure QI	15.00	15.12	15.02	15.05	1.71	1.67	1.71	1.70	25.59

2.2 Tensile Testing Procedure

The tensile test was conducted in accordance with standard BS EN ISO 527-5:2009 for unidirectional 0° and 90° specimens, as well as the $\pm$ 45° specimen [1]. BS EN ISO 527-5:1997 is the standard used when testing the 15°, 30°, and 45° specimens. Both cover the tensile testing of prepregs and carbon fibre composites.

For the 0° and 90° specimens, the initial distance between grips was set at 136 mm, and the end tabs were more than 50 mm in length, and between 0.5 mm and 2 mm thick, as per BS EN ISO 527-5:2009 [2]. Care was taken to ensure the pressure applied by the grips does not crush the specimen, only enough to

prevent slipping when under load. The end tabs were inserted at least 7 mm into the grips. The test speed was 2 mm per minute. The 0° specimen is thinner in the width and thickness dimensions to ensure tensile rather than grip failure.

The 15°, 30°, and 45° specimens, like the 0° and 90° specimens, are rectangular with end tabs, and are known as type 3 according to BS EN ISO 527-5:1997. The initial distance between tabs is 150 mm, and 136 mm between the grips. The test speed was 2 mm per minute.

White paint was applied to the front surface of each specimen, and dots were applied using a roller for the IMETRUM digital image correlation (DIC) camera to measure longitudinal and transverse strain, eliminating the need for strain gauges. The DIC camera was positioned level with the specimen, and a steel rule placed beside it in view for calibration. Vertical and horizontal strains are added along with a 2D map. The tensile test machine is the MTS Criterion Series 40 300 kN Universal Testing Machine. Before starting, readings are zeroed for all tests, and data were zeroed in the analysis where necessary.

2.3 Compression Test Procedure

The procedure for the compression test is outlined by standard ASTM D6641 [3]. The MTS Criterion Series 40 300 kN Universal Testing Machine with a combined loading compression (CLC) fixture was used, combining both end and shear loading. Specimen dimensions were as per Table 2, with tabs applied and a gauge length of 13 mm to prevent buckling.

A strain gauge of gauge length 2 mm was bonded to each specimen using a thin layer of adhesive. Great care was taken to align the specimen in the loading direction and ensure tabs were fully seating in the fixture. The CLC bolts were tightened in a diagonal pattern to 3 Nm using a torque wrench. Loading was applied at 1.3 mm per minute for both 0° and 90° specimens. After failure, the failure mode was compared to Figure 4 of ASTM D6641 to ensure acceptability.

2.4 Flexural Test Procedure

The flextural tests followed BS EN ISO 14125:1998 [4]. The three-point bending method was used with the Instron 3369 50 kN Universal Testing System. The specimen length was 100 mm, spanning 80 mm between the outer support points. As per Table 2, the rectangular specimens were approximately 15 mm wide and 2 mm thick. The load was applied to the top surface at a rate of 2 mm per minute.

3 Results and Discussion

3.1 Comparison of E_1, E_2, G_{12} of unidirectional lamina to Micromechanics Theory

The theoretical values of the longitudinal Young's modulus $E_{1,t}$ and transverse Young's modulus $E_{2,t}$ were calculated using the Rule of Mixtures equations. The data sheet for the T700G standard modulus carbon fibre gives a fibre modulus of E_f = 240 GPa [5]. The VTC401 Epoxy Component Prepreg data sheet does not specify the matrix modulus, but a typical value for epoxy is E_m = 2.4 GPa [6]. Assuming a fibre volume fraction of 55%, $E_{1,t}$ and $E_{2,t}$ can be calculated using Equation 1 and 2:

$$E_{1,t} = \phi_f E_f + (1-\phi_f)E_m = 0.55 \times 240 \text{ GPa } + (1-0.55) \times 2.4 \text{ GPa } = 133.08 \text{ GPa} \tag{1}$$

$$E_{2,t} = \frac{E_f E_m}{(1-\phi_f)E_f + \phi_f E_m} = \frac{240 \text{ GPa } \times 2.4 \text{ GPa}}{(1-0.55) \times 240 \text{ GPa } + 0.55 \times 2.4 \text{ GPa}} = 5.27 \text{ GPa} \tag{2}$$

The experimental longitudinal Young's modulus, $E_{1,e}$ is the slope of the stress-strain curve of the unidirectional 0° specimen, where tensile stress σ is plotted against the strain ϵ in the loading direction. Similarly, the experimental transverse Young's modulus, $E_{2,e}$ is obtained from the linear region of the stress-strain curve of the UD 90° specimen, minus the initial toe region. Figure 1 shows both curves.

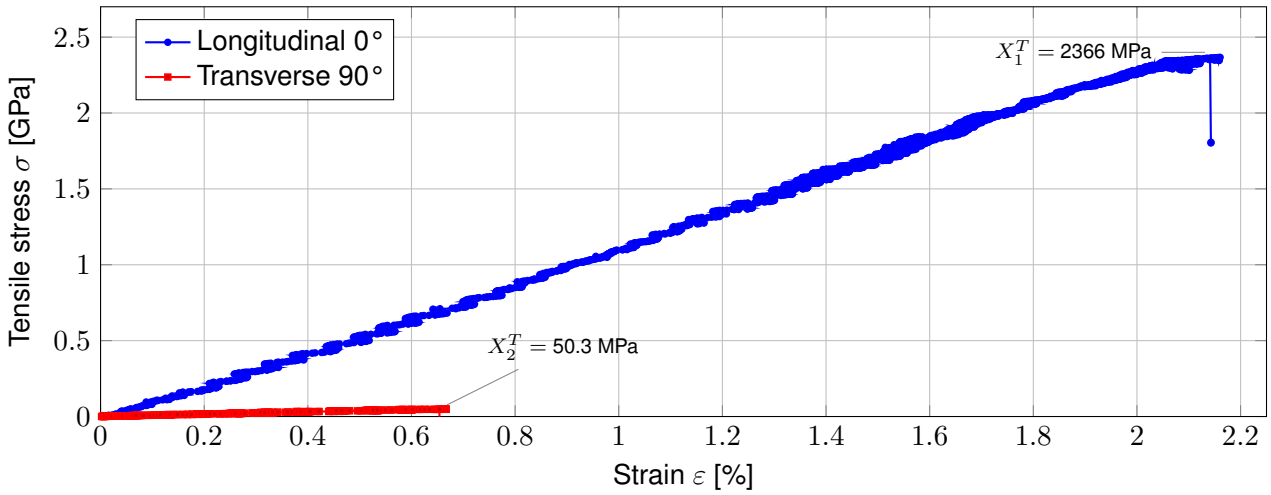


Figure 1: Stress–strain curves for longitudinal and transverse laminates

From the experimental data exported to Excel, stress was plotted against the strain in the direction of force application. For the 0-degree specimen, the curve is nearly linear until failure, indicating brittle failure of the fibres. Elastic deformation ended at ϵ = 0.0205 mm/mm and σ = 2.4 GPa, with failure at ϵ = 0.0216 mm/mm and σ = 2.45 GPa. The plastic region is excluded before applying the line of best fit. The slope gave $E_{1,e}$ = 117.9 GPa. For the 90-degree specimen, there is no clear plastic region. Failure occured at ϵ = 0.0065 mm/mm and σ = 48 MPa, giving $E_{2,e}$ = 7.3 GPa.

The in-plane shear modulus G_{12} can be determined theoretically if the fibre and matrix shear moduli, G_f and G_m are known. Literature gives G_f = 27 GPa, and G_m values range from G_m = 1.18 GPa to 2 GPa [7] [8]. Using Equation 3 with ϕ_f = 0.55 gives a range of $G_{12,t}$ = 2.49 - 4.08 GPa.

$$G_{12,t} = \frac{G_m G_f}{G_f(1-\phi_f) + G_m \phi_f} \tag{3}$$

$$G_{12,e} = \frac{\sigma}{2(\varepsilon_x - \varepsilon_y)} \tag{4}$$

Experimentally, G_{12} was obtained from the gradient of stress applied σ against $2(\epsilon_x - \epsilon_y)$, where ϵ_x and ϵ_y are the strains in the longitudinal and transverse direction, as shown in Equation 4, from the data of $\pm 45°$ specimen tensile test. Taking the slope of the linear section of the elastic region, $G_{12,e}$ = 3.07 GPa. The percentage difference is 17.9%. The experimental value was used for subsequent calculations as it uses data collected from the tested prepreg material.

A comparison between the theoretical values and experimentally derived values are shown in Table 4. The theoretical E_1 was 11.4% higher than the experimental. Fibre breakage during loading likely reduced the stiffness and caused the stress-strain curve to curve downwards. Layer misalignment, imperfect bonding, voids, and tab slippage could be contributing factors. The theoretical E_2 is known to be an underestimate due to the inhomogeneity of stresses within the matrix, 38.4% for this experimental data. The slab assumption is known to be poor; the Halpin–Tsai equations may give a more accurate result. Additionally, if E_m is underestimated, Equation 2 would exaggerate the discrepancy because the result if matrix dominated. The experimental G_{12} lies near the lower bound of the theoretical range, suggesting a lower G_m or fibre volume ϕ_f value.

Table 4: Comparison between theoretical and experimental elastic constants of the UD composite lamina

Property	**Symbol**	**Theoretical [GPa]**	**Experimental [GPa]**	**Difference [%]**
Longitudinal modulus	E_1	133.08	117.9	11.4
Transverse modulus	E_2	5.27	7.3	38.5
In-plane shear modulus	G_{12}	2.49 - 4.08	3.07	-

3.2 Comparison of X_1^T, X_2^T, S_6 of unidirectional lamina to data sheets

The experimental longitudinal tensile strength X_1^T was the maximum stress experienced by the unidirectional 0° specimen at failure on its stress-strain curve. By the same method, the experimental transverse tensile strength X_2^T was determined. These values are X_1^T = 2366 MPa and X_2^T = 50.3 MPa.

The in-plane shear strength S_6 was the maximum in-plane shear stress τ_6 experienced. Therefore, S_6 is the maximum stress value on the ±45° stress-strain curve, therefore S_6 = 120 MPa. This value is 6.3% lower than the data sheet value of 128 MPa [9].

Table 6 compares the experimental values to those from the T700G fibre data sheet [5]. The T700G data sheet assumes 60% fibre volume fraction, while the experimental laminate was assumed to be 55%. The data sheet epoxy was #2500 cured at 130°C. The experimental longitudinal tensile strength was 16% higher than the data sheet value, while the transverse strength was 32.9% lower, both consistent with fibre volume fraction lower than 60%, as expected. The experimental in-plane shear strength was 6.3% lower than the theoretical, likely due to voids or imperfect layer adhesion from manual layup. Minor strain or stress uncertainties are also possible, along with differences in epoxy type.

Table 5: Comparison of experimental and data sheet strength values for unidirectional laminate

Property	Symbol	Experimental [MPa]	Data sheet [MPa]	Difference [%]
Longitudinal tensile strength	X_1^T	2366	2040	16.0
Transverse tensile strength	X_2^T	50.3	75	32.9
In-plane shear strength	S_6	120	128	6.3

3.3 Comparison of E_x predictions of angle piles with off-axis experimental data

To predict the Young's modulus E_x for 15°, 30° and 45° ply orientations, Equation 5 is used:

$$\frac{1}{E_{x,t}} = m^4 \frac{1}{E_{1,t}} + m^2 n^2 \left(\frac{1}{G_{12,t}} - \frac{2\,\nu_{12}}{E_{1,t}} \right) + n^4 \frac{1}{E_{2,t}}, \qquad \text{where} \quad m = \cos\theta, \quad n = \sin\theta \tag{5}$$

Using $E_{1,t}$ = 133.08 GPa, $E_{2,t}$ = 5.27 GPa, ν_{12} = 0.29 and G_{12} = 3.07 from Section 3.1, the theoretical values were obtained, as shown in Table 6. Poisson's ratio ν_{12} was obtained using the rule of mixtures, with common values for the epoxy ν_m and fibres ν_f, 0.34 and 0.25, shown in Equation 6 [7].

$$\nu_{12} = \phi_f \nu_f + (1 - \phi_f)\nu_m = 0.55 \times 0.25 + 0.45 \times 0.34 \approx 0.29 \tag{6}$$

Experimental E_x values were taken from the slope of the elastic region of the tensile stress–strain curves for +15°, +30°, and +45° layups. For the +15° specimen, the slope was taken over the first 40%, due to the extent of the curvature. Figure 2 shows the stress-strain curves.

It can be seen in Table 6 that the experimental and theoretical values for the +15° angle differ by only 1.1%, 4.4% for +30° and 12.1% for the +45° specimen. The increasing deviation with angle reflects greater matrix influence and shear deformation effects at higher fibre orientations. As the fibre angle increases, the matrix carries a larger portion of the load, making results more sensitive to behaviour such as shear deformation and imperfect bonding. The slope of the stress-strain curve becomes smaller, making experimental E_x more sensitive to noise and strain gauge misalignment. Equation 5 assumes linear, homogeneous behaviour, an approximation that amplifies the effect of any inaccuracies in E_2, G_{12} and ν_{12} as the angle increases.

Table 6: Comparison of experimental and datasheet strength values for unidirectional laminate

Property	Symbol	Theoretical [GPa]	Experimental [GPa]	Difference [%]
+15° Modulus	$E_{x,\theta=15}$	35.7	35.3	1.1
+30° Modulus	$E_{x,\theta=30}$	13.6	14.2	4.4
+45° Modulus	$E_{x,\theta=45}$	8.58	7.54	12.1

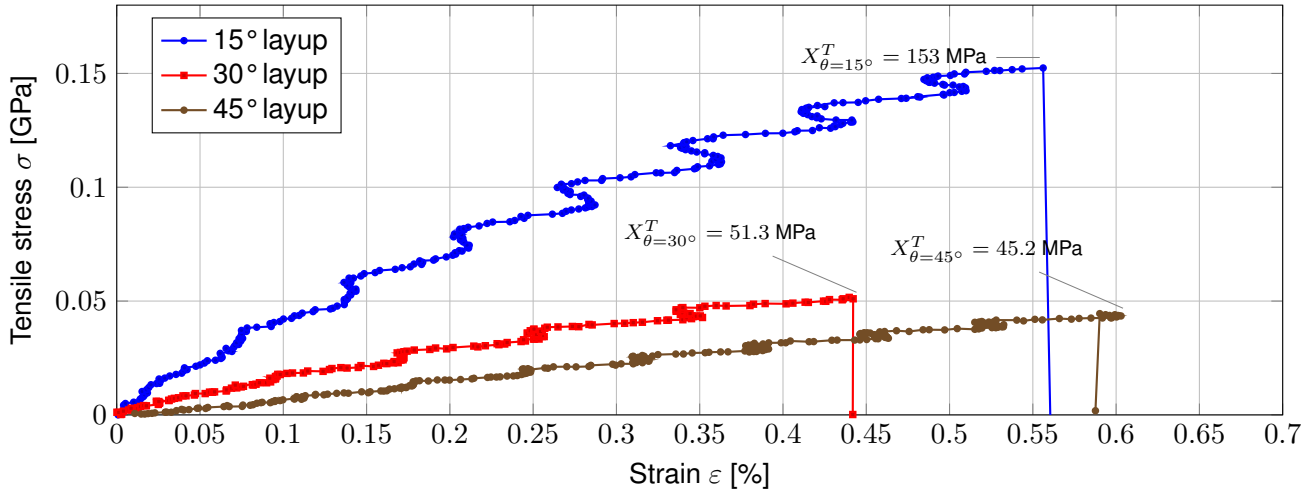


Figure 2: Stress–strain curves for 15°, 30°, and 45° laminates

3.4 Comparison of X^T of all tested laminae

Figure 3 shows the experimental tensile strength X^T for different fibre orientations. The results show an exponential decrease with increasing fibre angle relative to the load direction. The 0° lamina exhibited the highest strength of 2366 MPa, since the fibres aligned with the load and carried most of the applied stress.

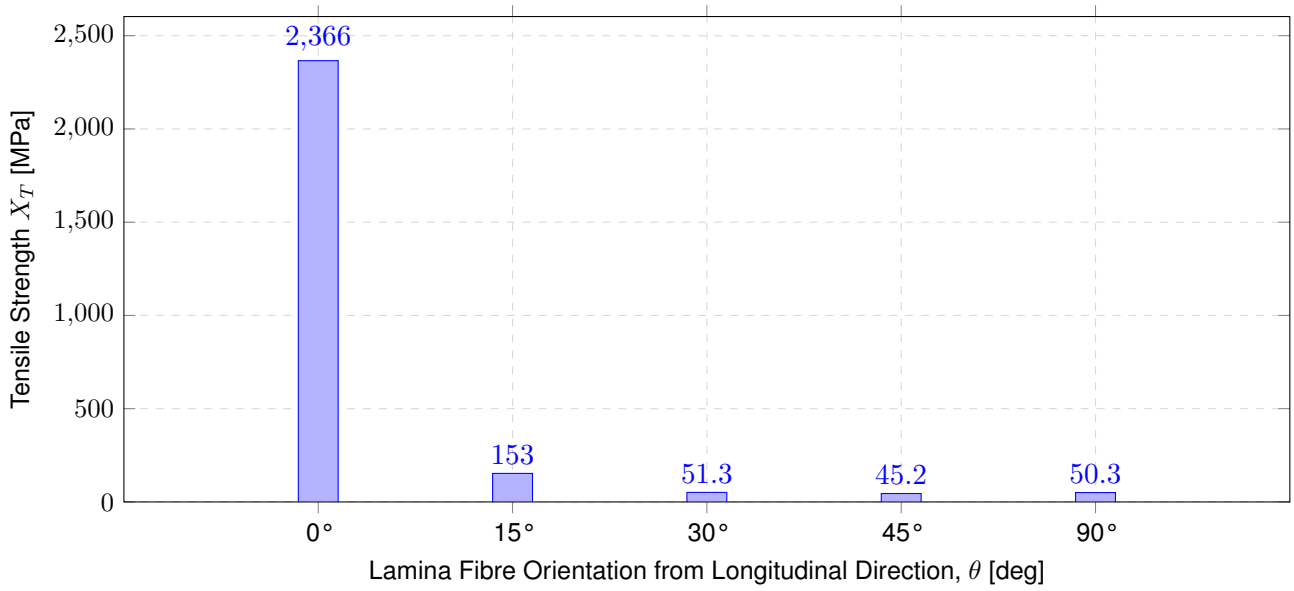


Figure 3: Comparison of experimental longitudinal tensile strength (X_T) for all tested laminae.

As the fibre orientation increased to 15°, 30°, 45°, and 90°, the tensile strength reduced sharply to below 200 MPa. This trend marks the transition from fibre-dominated failure at 0° to matrix and shear-dominated failure mechanisms at larger angles. Even at small angles, fibres no longer connect the end tabs and grips, forcing load transfer through the matrix and resulting in significantly reduced strength.

The strength reduction is caused by increased shear stresses within the matrix and at the fibre–matrix interface, causing premature cracking and debonding. These results are consistent with the expected anisotropic behaviour of unidirectional carbon-fibre epoxy composites, where the longitudinal tensile strength greatly exceeds the transverse and off-axis strengths. The tensile strength at 90° (50.2 MPa) was higher than at 45° (45.2 MPa), indicating that the epoxy matrix is stronger in tension than in shear. At 45°, the failure mechanism is shear failure and fibre-matrix interface debonding, while at 0° it involved brittle fibre fracture, matching similar experimental results [10].

In addition to the unidirectional tests, the ±45° specimen failed at 120 MPa, exhibiting greater strength than the unidirectional 45° laminate, but lower than the 15° specimen, which failed at 153 MPa.

3.5 Comparison of compressive strengths X_1^C and X_2^C to tensile strength X_1^T and X_2^T

Figure 4 shows the compression stress-strain curves for the unidirectional laminate in the longitudinal and transverse directions. Stress was derived by dividing the force applied by the initial cross-sectional area. Note that the area would increase throughout the test, but are assumed constant for simplicity. The compressive strengths are the peak stress values before failure: X_1^C = 1040 MPa, and X_2^C = 127 MPa. Figure 4 shows the laminate remained stronger and stiffer longitudinally than transversely. The loop back on the X_2^C results from matrix cracking and strain gauge relaxation.

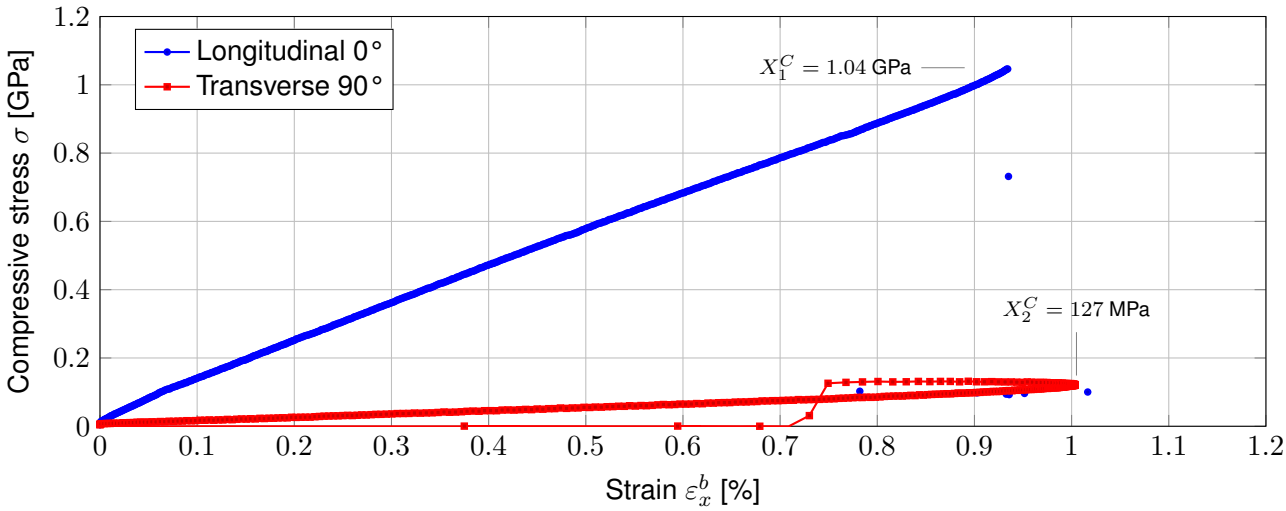


Figure 4: Compressive stress–strain curves for 0° and 90° laminates laminates

Table 7 compares compressive and tensile strengths (Section 3.2). The longitudinal compressive strength was 44% of the tensile strength, due to fibre micro-kinking and buckling in compression versus fibre-dominated load in tension [11]. In contrast, the transverse direction compressive strength is 2.5 times higher because crack propagation in tension is easier than matrix crushing in compression. These results show that the epoxy resin resists crushing more effectively than cracking, so the transverse compression test yields higher apparent strength. Imperfections in the fibre alignment or resin, such as voids, will exaggerate this.

Table 7: Comparison of compressive and tensile strengths in 0° and 90° laminates

Orientation	X^T [MPa]	X^C [MPa]	X^C/X^T
0°	2366	1040	0.44
90°	50.3	127	2.52

3.6 Comparison of flexural modulus E_x^b and strength σ_x^b of cross-ply and quasi-isotropic laminates

Data from the three-point bending tests showed the mid-span deflection and load applied. The span was 80 mm, with dimensions measured as depth d = 1.78 mm and width, b = 15.12mm for the cross-ply specimen and d = 1.70 mm and b = 15.05 mm for the quasi-isotropic specimen. The stress σ and flexural strain ϵ were calculated using Equations 7 and 8, where F is the load applied, L the span and D the mid-span deflection. The results are shown in Figure 5.

$$\sigma = \frac{3FL}{2bd^2} \tag{7}$$

$$\varepsilon = \frac{6Dd}{L^2} \tag{8}$$

Up to failure, both laminates showed elastic behaviour dominated by the fibres. The slope gave the flexural modulus, E_x^b, and the peak stress is the flexural strength, σ_x^b. These values are shown in Table 8 for the cross-ply and quasi-isotropic laminae. Figure 5 shows the flexural stress–strain responses.

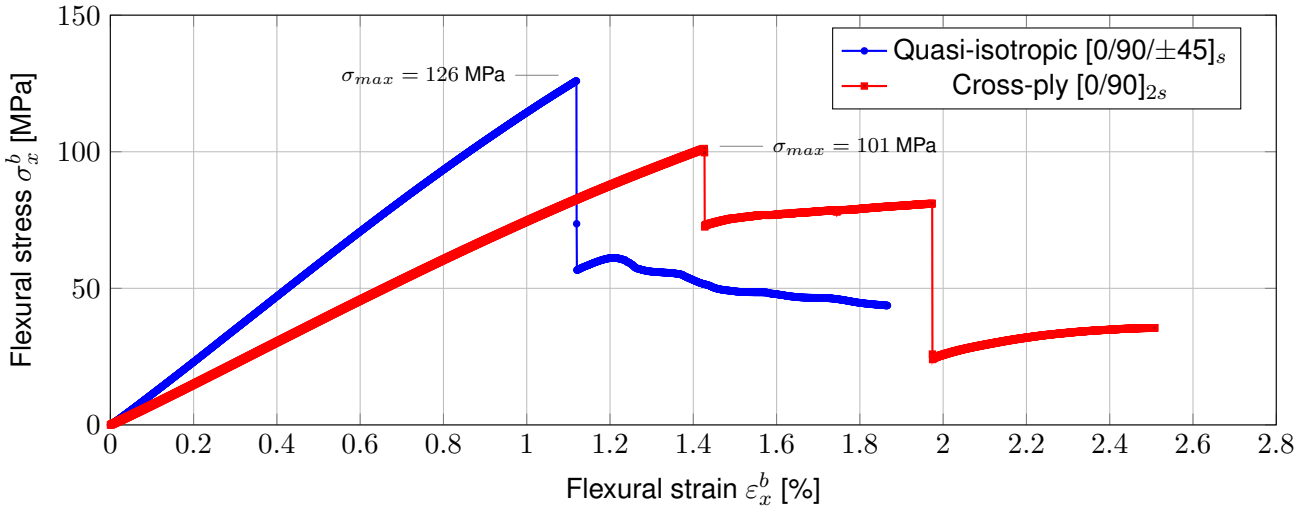


Figure 5: Flexural stress–strain curves for quasi-isotropic and cross-ply laminates

The quasi-isotropic laminate has a higher flexural modulus of E_x^b = 11.5 GPa and a higher flexural strength σ_x^b 126 MPa compared to the cross-ply laminate, where E_x^b = 7.63 GPa and σ_x^b = 101 MPa. This increase of 51% in stiffness and 25% in strength is due to the $\pm 45^\circ$ plies providing improved load distribution through the laminate thickness and better interlaminar shear resistance.

Both laminates experienced brittle failure, shown by the abrupt drop in stress post-peak. The cross-ply specimen, with only 0° and 90° plies, fails at a slightly higher strain (1.43%) than the quasi-isotropic laminate (1.11%). This suggests the cross-ply allows for some additional matrix-dominated deformation before stepped failure of layers, whereas the quasi-isotropic laminate fails earlier due to more efficient load transfer among plies.

Table 8: Comparison of flexural modulus E_x^b and flexural strength σ_x^b for cross-ply and quasi-isotropic laminates

Laminate Type	**Layup**	E_x^b **[GPa]**	σ_x^b **[MPa]**	**Failure strain [%]**
Quasi-isotropic	$[0/90/+45/-45]_s$	11.5	126	1.11
Cross-ply	$[0/90]_{2s}$	7.63	101	1.43
% Difference		50.7	24.8	22.4

3.7 Discussion of failure modes observed from all tests

Figure 6 and 7 show unidirectional tensile test failures after failure. The 0° tensile specimen failed due to brittle fibre fracture. The specimen broke into multiple fragments once the load-bearing fibres failed, as the load was transferred to the matrix, which subsequently failed due to its lower strength properties.

The 90° tensile specimen failed due to matrix cracking, as the fibres offer minimal structural support in this orientation. At 15°, 30°, and 45°, failure became increasingly matrix-dominated as the fibre angle increased. Shear stress within the matrix and at the fibre-matrix interface led to crack propagation along the fibre direction. Figure 7 shows evidence of delamination. Misaligned plies generate unequal shear forces and local stiffness variations, initiating microcracks that create delamination.

In the ±45° laminate, as shown in Figure 8, failure began along one ply angle first, likely the one oriented furthest from the longitudinal axis, producing interlaminar slippage and not total separation.

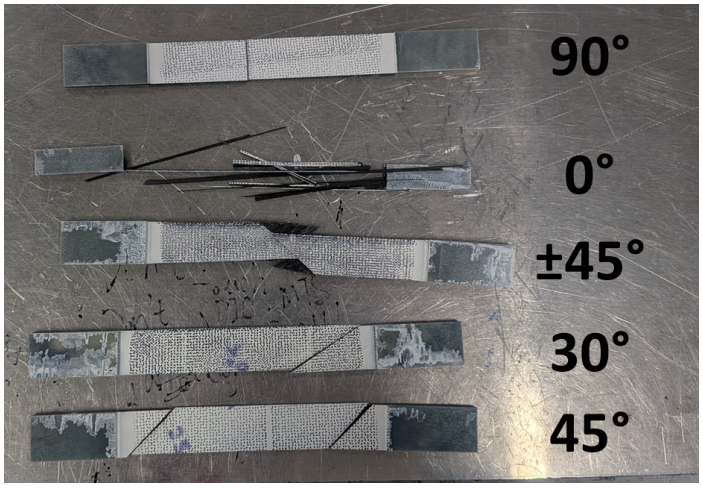


Figure 6: Tensile specimen failures.

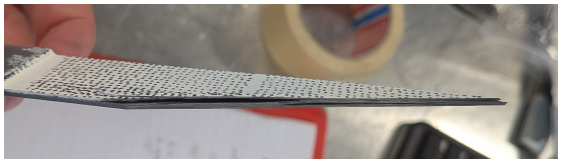


Figure 7: Delamination of a 15° specimen.

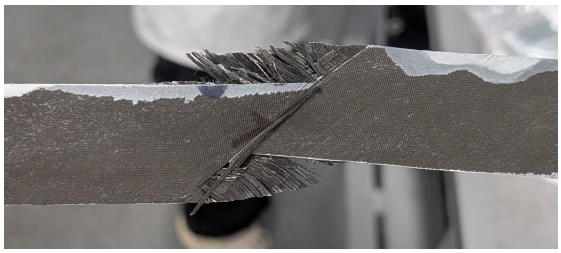


Figure 8: Failure of a ±45° specimen.

Under compression, the fibres in the 0° specimen failed by local micro-buckling and kinking. The specimen failed at 42.2% of the tensile failure load (Table 7). The fibres are thin (7 μm diameter [5]) and very stiff in tension but weak in compression, which does not effectively resist lateral deflection. As seen in Figure 9, failure has propagated through the matrix into the tabbing region. For the transverse 90° fibre orientation specimen, failure occurred through crushing of the matrix, which is less brittle and stronger than in tension.

Flexural loading (Figure 10 combines tension, compression, and interlaminar shear. The outermost plies experience the largest bending stresses as they are furthest away from the neutral axis, with tensile failure on the bottom and compressive failure on the top. The quasi-isotropic laminate failed abruptly due to compressive failure on the top surface, as the compressive strength of the laminate is lower than its tensile strength (Section 3.5). The cross-ply specimen failed progressively via multiple-layer cracks (see Figure 5) and interlaminar shear without total rupture at higher strain and lower stress.

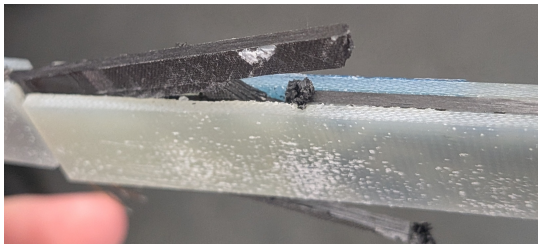


Figure 9: 0° specimen failing under compression

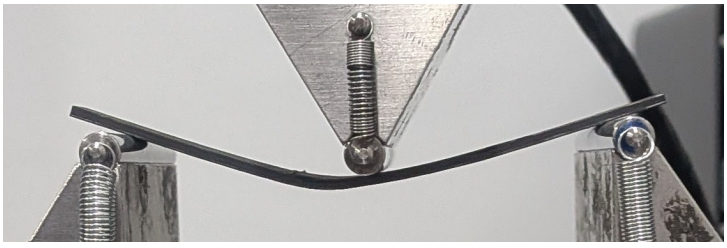
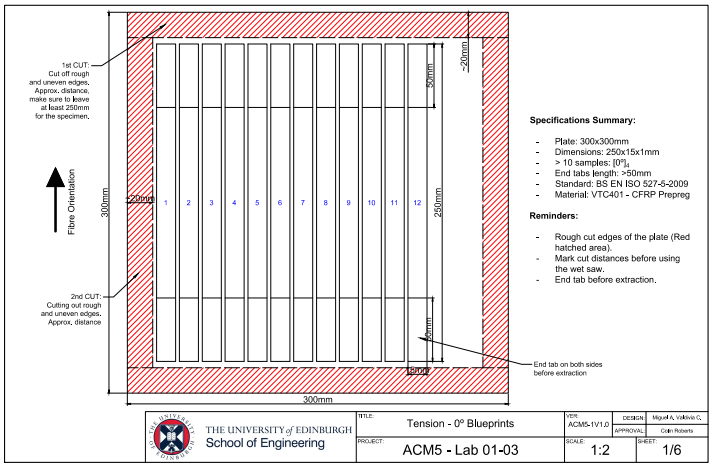
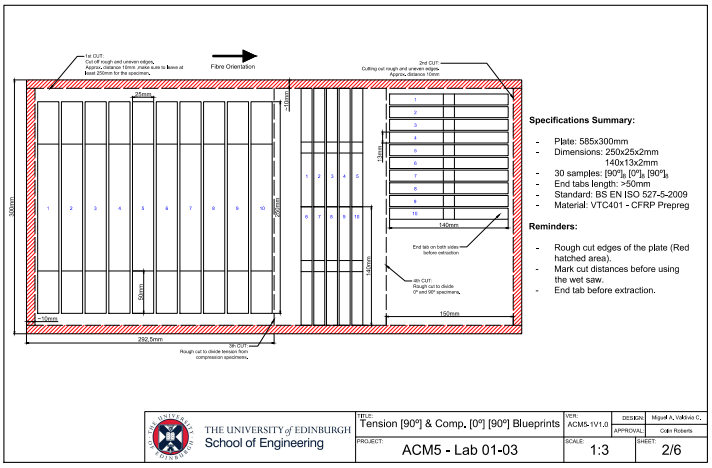
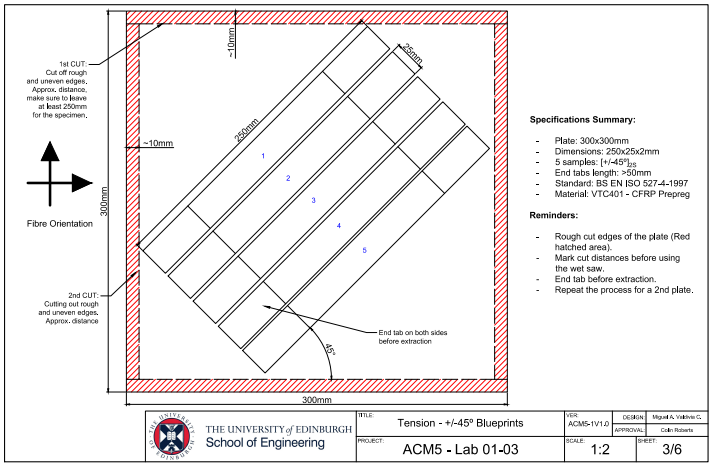


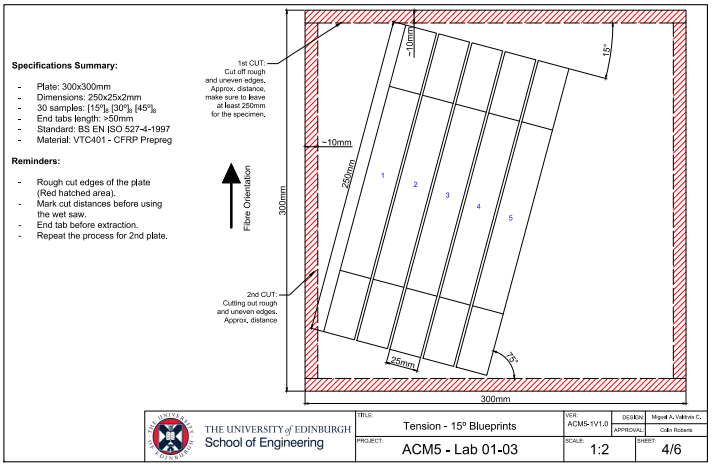
Figure 10: Specimen failure in flexure test, off-centre

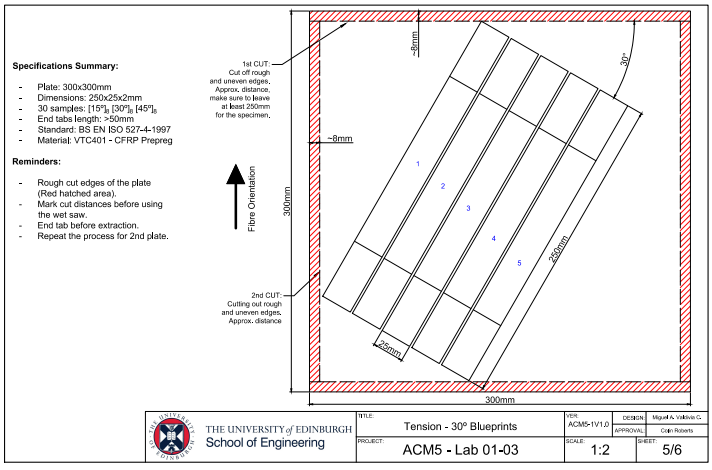
A Appendix - Specimen Extraction Drawings











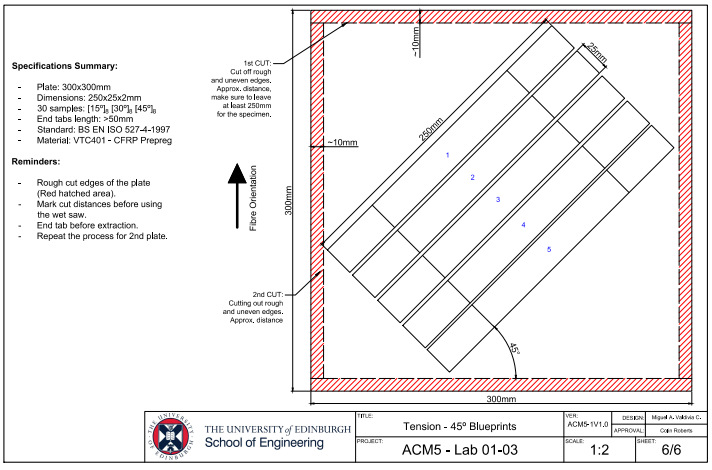


Figure 11: Specimen extraction drawings

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